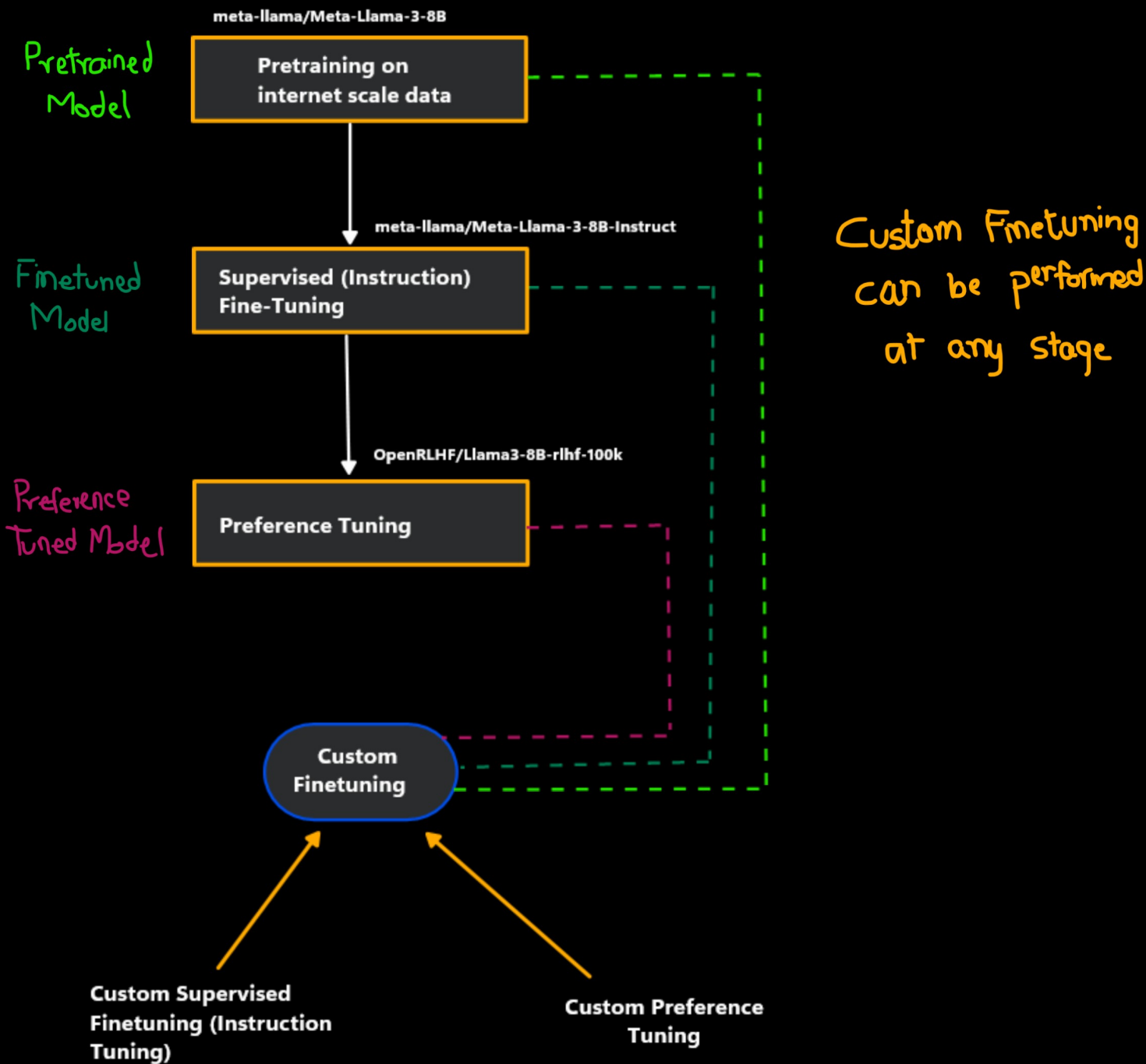


Fine-Tuning

Fine-tuning is the process of taking a already trained model and training it further on domain-specific or task-specific data so that it performs better for a particular use case

LLM Training Pipeline

- 1. Pretraining of a model
- 2. Supervised Finetuning (Instruction Finetuning)
- 3. Preference Tuning



Ways to Fine tune

- 1. Huggingface Modules
- 2. Unsloth

Methods of Finetuning

- 1. Full finetuning
- 2. Parameter Efficient Finetuning

① Pretraining ← Raw

② SFT/Instruction ← I/P ↔ O/P

③ Preference Tuning

Custom
Tuning

Train the model
on your own
data

① Pretraining (Unsupervised/Self supervised Learning)

② SFT → Supervised Finetuning

Parameter Level

① Full Finetuning X → Retrain all parameters

② Partial Finetuning → Subset of Parameters

Two methods

① Old school method

CNN
BERT
BART
TS

① Freeze all the layers and
train only last output layer

② Freeze some starting layers and
retrain some remaining last layers

② PEFT

Small Infrastructure

① LoRA/QLoRA

(Weight Decomposition) ② DoRA

③ Bitfit

④ IA³

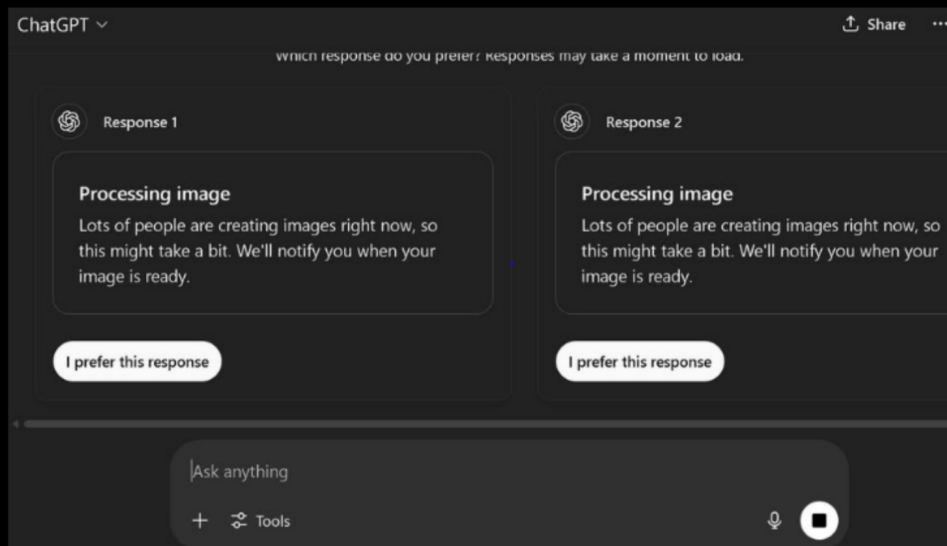
Quantized

float → int

↳ small float

X

③ Preference Alignment



① RLHF $\rightarrow$ OpenAI $\rightarrow$ RL $\rightarrow$ DRL $\rightarrow$ PPO

② GRPO

③ DPO

④ ORPO